

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 7: Ratio and Proportion

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Properties of proportion

1. Use the properties of proportion to solve $\frac{6x^2 + 3x - 5}{3x - 5} = \frac{9x^2 + 2x + 5}{2x + 5}$, where $x \neq 0$.

ICSE 2025, 3 marks

SOLUTION

Apply dividendo: if $\frac{a}{b} = \frac{c}{d}$ then $\frac{a-b}{b} = \frac{c-d}{d}$.

$$\frac{6x^2 + 3x - 5 - (3x - 5)}{3x - 5} = \frac{9x^2 + 2x + 5 - (2x + 5)}{2x + 5}, \text{ so } \frac{6x^2}{3x - 5} = \frac{9x^2}{2x + 5}$$

Since $x \neq 0$, divide both sides by $3x^2$: $\frac{2}{3x - 5} = \frac{3}{2x + 5}$

$$4x + 10 = 9x - 15, \text{ so } 5x = 25.$$

Answer: $x = 5$

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